

Mathematics Reference Guide

STEM High School for Boys - 6th of October

Algebra · Geometry · Trigonometry · Statistics

Authored by

STEM Mentorship

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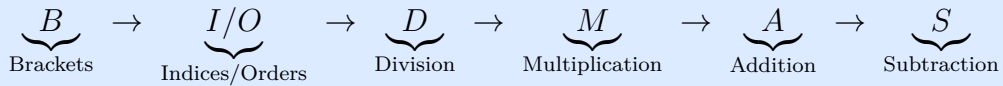
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Algebra

Order of Operations (BIDMAS / BODMAS)

Rule — BIDMAS / BODMAS

Evaluate expressions in this order:



Example

$$3 + 6 \times (5 + 4) \div 3 - 7 = 3 + 6 \times 9 \div 3 - 7 = 3 + 54 \div 3 - 7 = 3 + 18 - 7 = 14$$

Factoring & Factorization Types

Greatest Common Factor (GCF)

Factor out the largest common factor from all terms.

GCF

$$ax + ay = a(x + y)$$

Example

$$6x^2 + 9x = 3x(2x + 3)$$

Factor by Grouping

Grouping

$$ax + ay + bx + by = a(x + y) + b(x + y) = (a + b)(x + y)$$

Difference of Squares

Difference of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Perfect Square Trinomials

Perfect Squares

$$a^2 + 2ab + b^2 = (a + b)^2 \quad a^2 - 2ab + b^2 = (a - b)^2$$

Sum and Difference of Cubes

Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Completing the Square

Completing the Square

Given $ax^2 + bx + c$, rewrite as:

$$a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a}$$

Example

$$x^2 + 6x + 5 = (x + 3)^2 - 9 + 5 = (x + 3)^2 - 4$$

Special Factoring Formulas

Cube of a Binomial & Square of a Trinomial

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Solving Systems of Linear Equations

A system of two equations in two unknowns:

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

By Elimination

Multiply equations to make one coefficient equal; add or subtract to eliminate a variable.

By Substitution

Solve one equation for one variable and substitute into the other.

Graphical Interpretation

Lines	Solutions	Type
Intersecting	One unique solution	Consistent & independent
Parallel	No solution	Inconsistent
Coincident	Infinitely many	Consistent & dependent

Types of Numbers

Number Sets

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

- $\mathbb{N} = \{1, 2, 3, \dots\}$ — Natural numbers
- $\mathbb{W} = \{0, 1, 2, 3, \dots\}$ — Whole numbers
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ — Integers
- $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ — Rationals
- Irrationals: $\sqrt{2}, \pi, e, \dots$
- $\mathbb{R} = \mathbb{Q} \cup \{\text{Irrationals}\}$ — Real numbers

Perfect Squares & Cubes

n	n^2	n^3	$\sqrt[3]{n^3}$
1	1	1	1
2	4	8	2
3	9	27	3
4	16	64	4
5	25	125	5
6	36	216	6
7	49	343	7
8	64	512	8
9	81	729	9
10	100	1000	10

Functions

A **function** $f : A \rightarrow B$ assigns to each $x \in A$ exactly one $y \in B$.

Polynomial Functions

Common Types

- **Constant:** $f(x) = c$
- **Linear:** $f(x) = mx + c$
- **Quadratic:** $f(x) = ax^2 + bx + c, \quad a \neq 0$

Domain & Range

- **Domain:** all valid inputs x
- **Range:** all resulting outputs $f(x)$
- **Zeros:** values of x where $f(x) = 0$

For $f(x) = \frac{1}{x}$: Domain = $\mathbb{R} \setminus \{0\}$, Range = $\mathbb{R} \setminus \{0\}$.

Algebraic Fractional Functions

Rational Function

$$f(x) = \frac{P(x)}{Q(x)}, \quad Q(x) \neq 0$$

Domain excludes all x where $Q(x) = 0$.

Algebraic Fractions

Operations

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd} & \frac{a}{b} - \frac{c}{d} &= \frac{ad - bc}{bd} \\ \frac{a}{b} \times \frac{c}{d} &= \frac{ac}{bd} & \frac{a}{b} \div \frac{c}{d} &= \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc} \end{aligned}$$

Reduction: Cancel common factors in numerator and denominator.

Common Domain: Exclude all values making any denominator zero.

Sequences & Series

Geometric Sequences

Geometric Sequence

General term:

$$a_n = a_1 \cdot r^{n-1}$$

Sum of first n terms:

$$S_n = a_1 \cdot \frac{1 - r^n}{1 - r}, \quad r \neq 1$$

Exponential Growth and Decay

Exponential Model

$$y = a \cdot b^t, \quad b > 0$$

- $b > 1$: growth
- $0 < b < 1$: decay

Interest & Investment

Compound Interest

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

- A = final amount, P = principal, r = annual rate (decimal)
- n = compoundings per year, t = time in years

Exponential Decay (Depreciation)

$$V(t) = V_0(1 - d)^t$$

where d is the decay/depreciation rate.

Combinatorics

Permutations & Combinations

Permutation (order matters):

$$P(n, r) = \frac{n!}{(n - r)!}$$

Combination (order does not matter):

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n - r)!}$$

Example

From 10 students, choose a team of 3:

$$C(10, 3) = \frac{10!}{3! \cdot 7!} = \frac{10 \times 9 \times 8}{6} = 120$$

Geometry

Circles

Basic Circle Formulas

$$\text{Circumference} = 2\pi r \quad \text{Area} = \pi r^2$$

Arcs, Central & Inscribed Angles

Angle Theorems

- **Arc Length:** $\ell = r\theta$ (radians), or $\ell = \frac{\theta}{360} \cdot 2\pi r$ (degrees)
- **Central Angle** = Arc measure (same number of degrees)
- **Inscribed Angle** = $\frac{1}{2} \times$ Intercepted arc

Sector & Segment Areas

Sector & Segment

$$A_{\text{sector}} = \frac{\theta}{360} \pi r^2 \quad (\text{degrees})$$

$$A_{\text{segment}} = A_{\text{sector}} - A_{\Delta}$$

Cyclic Quadrilaterals

Cyclic Quadrilateral

Opposite angles are supplementary:

$$\angle A + \angle C = 180^\circ \quad \angle B + \angle D = 180^\circ$$

Ptolemy's Theorem: $AC \cdot BD = AB \cdot CD + AD \cdot BC$

Brahmagupta's Formula:

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}$$

Chord and Tangent Theorems

Chord Theorems

Intersecting Chords: If chords AB and CD intersect at P :

$$PA \cdot PB = PC \cdot PD$$

Tangent-Secant: From external point P :

$$(\text{tangent})^2 = \text{external} \times \text{whole secant}$$

Position of a Point Relative to a Circle

Let circle have centre O and radius r , and point P at distance $d = OP$:

- $d < r$: P is **inside** the circle
- $d = r$: P is **on** the circle
- $d > r$: P is **outside** the circle

Circumcircle & Incircle

Triangle Circles

Circumradius:

$$R = \frac{abc}{4K}$$

Inradius:

$$r = \frac{K}{s}, \quad s = \frac{a + b + c}{2}$$

where K is the area of the triangle.

Parallel Lines & Transversals

Angle Pairs with Parallel Lines

For lines $\ell_1 \parallel \ell_2$ cut by transversal t :

- **Corresponding angles:** equal
- **Alternate interior angles:** equal
- **Alternate exterior angles:** equal
- **Co-interior (consecutive interior):** supplementary $= 180^\circ$
- **Vertically opposite angles:** equal

Triangles

Angle Sum

$$\angle A + \angle B + \angle C = 180^\circ$$

Exterior angle of a triangle = sum of two non-adjacent interior angles.

Midsegment Theorem

The segment joining midpoints of two sides is parallel to the third side and half its length.

Pythagorean Theorem

Pythagorean Theorem

For a right triangle with legs a, b and hypotenuse c :

$$a^2 + b^2 = c^2$$

Euclidean Theorems (Right Triangle)

Altitude on Hypotenuse

If altitude h is drawn to hypotenuse $c = p + q$:

$$h^2 = p \cdot q \quad a^2 = p \cdot c \quad b^2 = q \cdot c$$

Triangle Inequality

$$|a - b| < c < a + b$$

(The sum of any two sides must be greater than the third side.)

Congruence Criteria

Criterion	Conditions
SSS	Three sides equal
SAS	Two sides and included angle equal
ASA	Two angles and included side equal
AAS	Two angles and non-included side equal
RHS	Right angle, hypotenuse, one side

Similarity Criteria

AA, SAS, SSS (proportional sides).

Area Formulas

Triangle Area

$$A = \frac{1}{2}bh \quad A = \frac{1}{2}ab \sin C \quad A = \sqrt{s(s-a)(s-b)(s-c)} \text{ (Heron's)}$$

where $s = \frac{a+b+c}{2}$.

Triangle Centers

- **Centroid:** intersection of medians; divides each median 2 : 1 from vertex.
- **Orthocenter:** intersection of altitudes.

- **Circumcenter:** intersection of perpendicular bisectors; equidistant from all vertices.
- **Incenter:** intersection of angle bisectors; equidistant from all sides.

Quadrilaterals & Polygons

Sum of Interior Angles of a Polygon

$$S = (n - 2) \times 180^\circ$$

where n is the number of sides.

Shape	Area	Key Properties
Square	a^2	All sides equal, all angles 90°
Rectangle	$l \times w$	Opposite sides equal, all angles 90°
Parallelogram	$b \times h$	Opposite sides $\parallel$ and equal
Rhombus	$\frac{d_1 d_2}{2}$	All sides equal, diagonals bisect perpendicularly
Trapezium	$\frac{(a+b)}{2} h$	One pair of parallel sides
Kite	$\frac{d_1 d_2}{2}$	Two pairs of adjacent equal sides

Coordinate Geometry

Key Formulas

Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope-Intercept Form:

$$y = mx + c$$

Point-Slope Form:

$$y - y_1 = m(x - x_1)$$

Parallel lines: $m_1 = m_2$ **Perpendicular lines:** $m_1 m_2 = -1$

Transformations

Rules

- **Translation by (h, k) :** $(x, y) \mapsto (x + h, y + k)$
- **Rotation by 90° CCW about origin:** $(x, y) \mapsto (-y, x)$
- **Rotation by 180° :** $(x, y) \mapsto (-x, -y)$
- **Reflection over x -axis:** $(x, y) \mapsto (x, -y)$
- **Reflection over y -axis:** $(x, y) \mapsto (-x, y)$
- **Reflection over $y = x$:** $(x, y) \mapsto (y, x)$

Volume & Surface Area of 3D Shapes

Shape	Volume	Lateral SA	Total SA
Cube (side a)	a^3	$4a^2$	$6a^2$
Cuboid	lwh	$2(lh + wh)$	$2(lw + lh + wh)$
Cylinder	$\pi r^2 h$	$2\pi r h$	$2\pi r(r + h)$
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r l$	$\pi r(r + l)$
Sphere	$\frac{4}{3}\pi r^3$	—	$4\pi r^2$
Pyramid (sq base)	$\frac{1}{3}a^2 h$	$2al$	$a^2 + 2al$
Prism	$A_{\text{base}} \times h$	$P_{\text{base}} \times h$	$2A_b + P_b h$

Note: l denotes slant height for cone and pyramid.

Trigonometry

Angles & Radians

Conversion Formulas

$$\text{Radians} = \frac{\pi}{180^\circ} \times \text{Degrees} \quad \text{Degrees} = \frac{180^\circ}{\pi} \times \text{Radians}$$

Degrees	Radians	Decimal (approx.)
0°	0	0
30°	$\pi/6$	0.524
45°	$\pi/4$	0.785
60°	$\pi/3$	1.047
90°	$\pi/2$	1.571
180°	π	3.142
270°	$3\pi/2$	4.712
360°	2π	6.283

Coterminal Angles

Two angles are coterminal if they share the same terminal side.

$$\theta_{\text{coterminal}} = \theta + 360^\circ \cdot k, \quad k \in \mathbb{Z}$$

Trigonometric Functions

Definitions in a Right Triangle

For angle θ in a right triangle with sides *opposite* (o), *adjacent* (a), *hypotenuse* (h):

SOH-CAH-TOA

$$\begin{array}{lll} \sin \theta = \frac{o}{h} & \cos \theta = \frac{a}{h} & \tan \theta = \frac{o}{a} \\ \csc \theta = \frac{h}{o} & \sec \theta = \frac{h}{a} & \cot \theta = \frac{a}{o} \end{array}$$

Reciprocal & Quotient Identities

Identities

$$\begin{array}{lll} \csc \theta = \frac{1}{\sin \theta} & \sec \theta = \frac{1}{\cos \theta} & \cot \theta = \frac{1}{\tan \theta} \\ \tan \theta = \frac{\sin \theta}{\cos \theta} & \cot \theta = \frac{\cos \theta}{\sin \theta} & \end{array}$$

Pythagorean Identity

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

ASTC Rule (All Students Take Coffee)

Signs of trig functions by quadrant:

Quadrant	Angles	sin	cos	tan
I	0°–90°	+	+	+
II	90°–180°	+	–	–
III	180°–270°	–	–	+
IV	270°–360°	–	+	–

Exact Values

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

Applications — Solving Right Triangles

Given any two elements (side or angle, apart from the right angle), find all others using:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}, \quad a^2 + b^2 = c^2$$

Statistics & Data

Mean, Median, Mode, Range

Definitions

For data set $\{x_1, x_2, \dots, x_n\}$:

$$\text{Mean : } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Median: middle value when data is ordered.

Mode: most frequently occurring value.

Range: $x_{\max} - x_{\min}$

Average Speed Formula

Average Speed

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

For two equal distances at speeds v_1 and v_2 :

$$\bar{v} = \frac{2v_1v_2}{v_1 + v_2}$$

Weighted Averages

Weighted Mean

$$\bar{x}_w = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Probability

Basic Probability

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}, \quad 0 \leq P(E) \leq 1$$

Complementary Events

Complement Rule

$$P(E') = 1 - P(E)$$

Mutually Exclusive Events

Events A and B cannot occur at the same time:

Addition Rule (Mutually Exclusive)

$$P(A \cup B) = P(A) + P(B)$$

Non-Mutually Exclusive Events

General Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Independent Events

Events A and B do not affect each other:

Multiplication Rule (Independent)

$$P(A \cap B) = P(A) \cdot P(B)$$

Dependent Events

Conditional Probability

$$P(A \cap B) = P(A) \cdot P(B | A)$$

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

Example — Deck of Cards

Probability of drawing two Aces without replacement from a 52-card deck:

$$P(\text{Ace}_1 \cap \text{Ace}_2) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$$